

Where to Apply: Mask-Guided Localization of LoRA Adapter Effects in Diffusion Models

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Abstract

Modern text-to-image generative models are powerful creative tools, and LoRA adapters have become a lightweight, widely-used way to add controllable edits on top of them. As low-rank weight updates, however, LoRA adapters act globally: in multi-subject scenes they steer every entity at once and leak into unrelated regions, dissolving the user’s localized intent. We propose a mask-guided application method that confines the effect of a globally-trained LoRA adapter to a chosen region, gating the adapter at inference time on an externally-provided segmentation mask through a per-step blend of the masked and unmasked predictions. We stress-test the method on Concept Sliders [4], continuous-scale LoRA adapters that are a particularly demanding case because they encode fine-grained semantic attribute directions rather than whole subjects, and introduce region-aware localization metrics built on CLIP and LPIPS to quantify where each edit lands. Our investigation spans both SDXL and FLUX.1-dev backbones. Code is available on [GitHub](#).

1. Introduction

Modern text-to-image generative models have become the de facto creative backbone for image synthesis. On top of a pretrained generator, *LoRA adapters* [7] have become a standard, lightweight way to add a specific controllable edit: a small low-rank update of selected weights, trained once and reused as a plug-in modulation of the model’s behaviour. LoRA adapters, however, act *globally*: an adapter trained to impose a specific identity, add a smile, increase age, or shift the visual style modulates every element of the generated image, regardless of whether the user wanted to edit it. This is harmless when the scene contains a single subject and the targeted attribute is well-localized to it, but it breaks down whenever the user’s intent is itself *localized*. Suppose we have generated a multi-subject scene we are happy with and only one of the subjects should appear older; or we want to change one person’s face with an identity LoRA while the scene contains several people; or

a landscape in which we want a painterly style only on the sky and not on the rest of the image. In all of these cases, applying a vanilla adapter produces *semantic leakage* by construction, leaving the localized intent without any mechanism to enforce it. The same low-rank modulation that makes LoRA adapters so practical, in other words, makes them blunt instruments in any generated scene that contains more than one element targeted by the adapter. In this work we localize the effect of a LoRA adapter, confining it to a chosen subject or region of a multi-subject scene. We propose a *mask-guided application* (Sec. 4) that leaves the adapter global and gates its effect, at inference time, on an externally-provided segmentation mask; toward its design we also considered an internal-mask alternative, which we discuss in Sec. 4. The mechanism applies unchanged to any scaled LoRA adapter, including identity or subject adapters. As our concrete test case throughout this work we use *Concept Sliders* [4], which train each adapter as a reusable, continuous knob over an interpretable attribute of the generated image, beyond what prompt engineering alone can deliver. They are a deliberately demanding probe for localization: each slider encodes a fine-grained semantic direction rather than a whole object, so an edit that leaks even slightly is immediately visible, and its continuously tunable strength lets us test localization across a range of edit intensities rather than at a single setting. To retain full experimental control over the attributes under study, we train our own Concept Sliders rather than relying on off-the-shelf checkpoints, and we run the investigation on two backbones from different families, SDXL and FLUX.1-dev. Throughout this work we focus on the editing of *generated* images, in which the adapter is applied directly during synthesis. The same local editing applies unchanged to *real* images, modulo a preliminary inversion step whose quality determines whether the resulting edit is useful. We verified this on a small set of images using *tight inversion* [8], but keep inversion out of our main experimental loop since its quality is an orthogonal variable to the localization mechanism.

2. Related Work

Image editing with diffusion models. Image editing with pretrained diffusion models has been pursued along several non-LoRA directions, including attention-based control such as *Prompt-to-Prompt* [5], model-inferred edit masks such as *DiffEdit* [2], and full-model subject-personalization recipes such as *DreamBooth* [13]. Our investigation instead focuses on the complementary line of LoRA-based editing.

LoRA-based editing. LoRA adapters [7] are widely used as continuous editing operators at inference time; *Concept Sliders* [4], our main test case, are one such family, trained as continuous attribute knobs and originally applied to single subjects. Multi-LoRA approaches have added spatial routing on top of LoRA: *LoRASHop* [3] performs region-controlled generation and editing on FLUX-style rectified-flow transformers, deriving subject priors from cross-attention to route per-LoRA contributions across non-overlapping regions. We instead obtain the localization mask externally, via a separate segmentation pass on the unedited generation (Sec. 4).

3. Preliminaries

We briefly review the two components on which our investigation rests: the generative backbones (Sec. 3.1) and the LoRA adapters our method operates on, of which Concept Sliders [4] are our test case (Sec. 3.2).

3.1. Generative backbones

We work with two text-to-image generators from two distinct families. *Stable Diffusion XL* [11] is a latent diffusion model [6] whose text-conditioned UNet predicts the noise $\varepsilon_\theta(x_t, t, c)$; *FLUX.1-dev* [1] is a rectified-flow [10] transformer whose text-conditioned MM-DiT predicts a velocity field $v_\theta(x_t, t, c)$. Both are latent models with attention and feed-forward layers amenable to LoRA injection, which is what lets a single LoRA formulation apply across both. They differ in the resolution at which the prediction lives: the SDXL UNet operates directly on a 128×128 latent, whereas the FLUX MM-DiT packs the latent into a 64×64 grid of 2×2 patches before transformer processing. This difference is what the spatial-mask formulation of Sec. 4 must account for, matching the mask to each backbone’s prediction grid.

3.2. LoRA adapters and Concept Sliders

A *LoRA adapter* [7] is a low-rank update of selected weight matrices of a frozen base model. Each adapted linear layer replaces its weight W with $W + \Delta W$, where $\Delta W = BA$ is constrained to low rank, so that only the small factors A and B are trained while W stays fixed. Originally introduced for parameter-efficient fine-tuning, LoRA adapters are now

widely used as plug-in modulations of a diffusion model’s denoising network: an adapter is trained once for a specific behaviour and attached at inference without retraining the base model. A *Concept Slider* [4] is a LoRA adapter specialized in two ways. First, it is trained to encode a single interpretable attribute (such as *age* or *smile*) as one direction in weight space. Second, it carries a continuous scalar scale $s \in \mathbb{R}$: each adapted layer applies $W' = W + s \cdot \Delta W$, so the LoRA-modulated noise predictor is ε_{θ^*} with $\theta^* = \theta + s \theta_s$, and s controls the strength of the edit at inference ($s = 0$ recovers the unmodulated model, positive s expresses the attribute more strongly and negative s its opposite). Each slider is trained with a *prompt-based* recipe that supervises the low-rank update from paired text prompts describing a target concept with and without the attribute, so that the learned direction pushes the output toward the attribute and away from its negation. We use Concept Sliders as the test adapters throughout this work, for two reasons. First, being LoRA adapters, they are a direct instance of what the localization method of Sec. 4 operates on. Second, they are a stringent and informative probe for it: their fine-grained attribute direction makes any leakage onto the wrong region immediately visible, and their continuous scale lets us apply the very same adapter at a range of strengths, so we can verify that localization holds not at a single operating point but across the whole edit-strength range that an editing and control tool would actually use. We train our own sliders (hyperparameters in Sec. 5.2) and use each one as a fixed adapter: the method leaves the adapter weights untouched and acts only at inference time.

4. Methodology

Our method takes a globally-trained LoRA adapter and localizes its effect spatially through a three-stage inference pipeline. We first *generate* an unedited image with the base model. We then *segment* that image externally (using SAM [9] in our experiments, though the choice of segmentation method is orthogonal to the mechanism and any model returning a sufficiently accurate region of interest can be substituted) to obtain a mask over the target subject(s). We finally *re-generate* with the same metadata (same prompt, same random seed, same denoising schedule), applying one or more adapters gated by the mask through the operator defined below. The mechanism is backbone-agnostic and we apply it to both SDXL and FLUX. Although we demonstrate it on Concept Sliders, it applies to any LoRA adapter with a continuous scale.

Edit operator. The re-generation pass runs in two phases, following a late-step activation schedule for the LoRA adapter [4]. In the high-noise portion of the trajectory, up to a threshold t^* , we run the base model alone, so that the global composition of the scene (such as subject positions and overall image structure) is not disturbed while it is be-

ing formed. Past t^* , at each denoising step we run two forward passes of the backbone (one with the adapter disabled, one with it active) and blend their predictions through the user-supplied mask, before passing the result to the scheduler. Formally, let $u_\theta(z_t, c, t)$ denote the model’s prediction at step t , ε_θ for SDXL, v_θ for FLUX (Sec. 3.1). Given the binary mask M , downsampled to the backbone’s prediction resolution, and an adapter with parameters $\Delta\theta$ at scale s , the spatial blend computed at each step of the active phase is

$$\hat{u}_t = (1 - M) u_\theta + M u_{\theta+s\Delta\theta}, \quad (1)$$

fed to the scheduler in place of u_θ . The adapter thus steers the trajectory only inside the masked region; outside, the model continues to follow its unmodified prediction. Because the blend is applied at every denoising step, the two forwards at step t both start from the same latent z_t (produced by the scheduler from the previous step’s blended prediction), so the trajectory inside and outside the mask evolves from a shared starting point at each step, which keeps the two sides coherent and prevents seam artefacts at the mask boundary. The operator extends naturally in two directions. (i) Multiple disjoint masks $\{M_i\}_{i=1}^N$, each gated by its own adapter $\Delta\theta_i$ at scale s_i , are handled by summing over the masked regions in Eq. (1), at a cost of $1+N$ forward passes per step. (ii) Multiple adapters inside the same masked region reduce to the standard additive aggregation of LoRA updates, $s\Delta\theta = \sum_j s_j \Delta\theta_j$, and incur no additional forward pass.

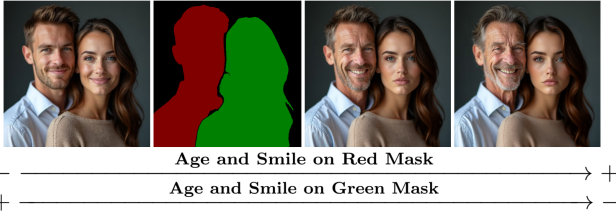


Figure 1. An example of the external mask-guided application, generated with FLUX: two disjoint masks, one per subject, each gated by two LoRA adapters (*age* and *smile*). From left to right: the unedited generation, the two masks, and the localized edits. Further examples are provided in Sec. C.

An internal-mask alternative we explored. We additionally adapted LoRASHop [3] (a training-free multi-LoRA pipeline whose mask is derived internally from the model’s own cross-attention). Our adaptation threaded a per-adapter continuous scale s_i through LoRASHop’s residual-blending machinery, so that each adapter’s contribution inside its region is weighted by its own strength rather than by the default toggled-on value, and allowed multiple adapters to compose additively inside the same region. The internal cross-attention mask, however, proved

too imprecise for fine-grained attribute edits, which is what motivated the external-mask design described above. An example of the extracted internal masks is provided in Fig. 14.

5. Experiments and Results

5.1. Metrics

To evaluate edits in a localized way, we work along two axes, perceptual change (LPIPS) and semantic alignment with the attribute prompt (CLIP), adapting each standard metric into a region-aware variant that captures both the change inside the targeted region and the preservation outside it. Let I_0 denote the unedited image, I_1 the edited image, and let m_a, m_b be a pair of disjoint binary masks defining the two regions being compared (with A_x the area of region x). We define two general scores parameterized by the choice of region pair. The *modified LPIPS* score normalizes the LPIPS change by area in each region and takes their ratio:

$$\text{LPIPS}_{ab} = \frac{d_a}{d_b + \varepsilon}, \quad d_x = \frac{\text{LPIPS}(I_0 m_x, I_1 m_x)}{A_x + \varepsilon}. \quad (2)$$

The *modified CLIP* score measures what fraction of the attribute-aligned change in CLIP text–image similarity is concentrated in region a , relative to the total change over the two compared regions:

$$\text{CLIP}_{ab} = \begin{cases} 0 & \text{if } \Delta_a \leq 0 \\ \frac{\Delta_a}{\Delta_a + |\Delta_b| + \varepsilon} & \text{otherwise,} \end{cases} \quad (3)$$

where $\Delta_x = C(I_1 \star m_x) - C(I_0 \star m_x)$ and the operator $\star$ keeps the pixels of the image inside m_x and replaces those outside with uniform mid-gray (RGB 127). This neutral background lets CLIP evaluate the region of interest in a plausible full-image context, without picking up signal from outside the masked area. The two scores have different ranges: $\text{CLIP}_{ab} \in [0, 1]$, with values close to 1 meaning that the attribute-aligned CLIP gain is concentrated almost entirely in region a , and values near 0.5 indicating a comparable gain in the two regions. LPIPS_{ab} , instead, is an unbounded ratio: a value of k means that the area-normalized perceptual change in region a is k times larger than in region b , and a value near 1 indicates comparable perceptual change across the two regions. For our tasks, in both metrics higher is better.

Localization metrics ($\text{LPIPS}_{\text{LOC}}$, CLIP_{LOC}). We instantiate the two general scores above as a pair of localization metrics, used throughout the experiments of Sec. 5.3. We set $m_a = m$ (inside the target mask) and $m_b = \bar{m}$ (outside); high values mean the edit concentrates inside the mask region.

Implementation. We compute LPIPS with the AlexNet backbone [14] and CLIP with the OpenAI CLIP-ViT-B/32 model [12]. We set $\varepsilon = 10^{-8}$ throughout. Every metric value reported in this paper is the *median* over the evaluation scenes, not the mean. There are two reasons. First, $\text{LPIPS}_{\text{LOC}}$ is an unbounded ratio: it is naturally right-skewed, and prone to inflation from individual scenes where the denominator (the area-normalized perceptual change in the comparison region) is close to zero. Second, although CLIP_{LOC} is bounded in $[0, 1]$, it often distributes bimodally between successful edits and edits that miss the targeted attribute. The empirical coefficient of variation across scenes is in the 0.6–0.8 range for the LPIPS score, so the median is a markedly more robust summary of the typical edit than the mean.

5.2. Experimental plan

We do not rely on any external dataset: all sliders and all evaluation scenes are obtained from hand-authored prompts and fixed seeds, all included in the code repository.

Backbones and training. We work with two text-to-image backbones at 1024×1024 image resolution: Stable Diffusion XL base 1.0 [11] and FLUX.1-dev [1], each used with its standard inference configuration (50 Euler steps with classifier-free guidance scale 7.5 on SDXL; 30 flow-matching steps with distilled guidance 3.5 on FLUX). The LoRA adapters used as test cases are Concept Sliders trained with the prompt-based recipe of Sec. 3.2. On SDXL we used a single training configuration for every slider, since the prompt-based Concept Sliders recipe was originally developed on Stable Diffusion models and trains reliably there: each slider is a rank-4 LoRA ($\alpha = 1$) injected into the UNet’s self-attention and convolutional layers but not its cross-attention, optimized with AdamW at a constant learning rate of 2×10^{-4} for 1000 iterations in bfloat16. FLUX, a rectified-flow transformer the recipe was not designed for, did not train stably out of the box and required a hyperparameter search: we swept the LoRA rank over $\{4, 16\}$, the injection site between cross-attention only and the broader non-cross-attention set, the learning rate over $\{2 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$, the schedule length over $\{300, 500, 1000\}$ steps, and the warm-up over $\{100, 200\}$ steps. The configuration adopted for the reported FLUX sliders, a rank-16 LoRA with non-cross-attention injection trained with AdamW at 2×10^{-4} for 1000 steps and a 100-step warm-up, was the one that trained most consistently across attributes.

Slider portfolio. We start from a pool of Concept Sliders covering six attributes (*age*, *smile*, *curly hair*, *fur length*, *painterly style*, and *day/night*), all trained globally with the classical Concept Sliders recipe [4]. This full set of six is trained on SDXL; on FLUX we omit *day/night*, leaving five sliders.

Evaluation scenes and masks. For each experiment we generate 20 multi-subject scenes per slider, with fixed seeds 1001–1020 and prompts hand-authored per backbone. These prompts feature more subjects in everyday settings (e.g., a couple at a Parisian café, a parent and a child baking in a kitchen, a pair of dogs in a meadow) for the subject-bound attributes, and single foreground subjects against a clear background (a sailing boat on the sea, a windmill in a field) for the style attributes. This yields $6 \times 20 = 120$ scenes for the evaluation on SDXL and $5 \times 20 = 100$ scenes for the same evaluation on FLUX. The masks used to evaluate localization are the same SAM masks produced at stage 2 of the three-stage pipeline (Sec. 4) and applied at inference.

Mask source and inference-time parameters. All subject masks in this work are produced with SAM [9] using the ViT-H checkpoint, prompted with a single positive point click on the subject. For the mask-guided application, we sweep the slider scale over three values per concept, chosen to span a useful effect range for that attribute (e.g., $\{0.5, 1.0, 1.5\}$ for facial attributes, and larger values for the global style sliders, such as $\{3, 4, 5\}$ for *day/night*). We refer to them as scale 1, scale 2, scale 3 throughout the tables. The late-step activation threshold (Sec. 4) is held fixed across attributes for a fair comparison: in both backbones, the blend activates after roughly the first 25% of the denoising trajectory (concretely, at timestep 750 on SDXL, whose schedule counts down from 999 to 0, and at denoising step 8 on FLUX, whose schedule counts up from 0 to 30).

Baseline. As a baseline, we apply the same sliders globally, without spatial masking, on the same scenes and scale ranges.

5.3. Results: External Mask-Guided Application

We evaluate the mask-guided application on the six SDXL concepts (120 scenes) and the five FLUX concepts (100 scenes), with localization computed against the SAM mask used at inference (complete results in Sec. A). Across both backbones the localization is strong: CLIP_{LOC} exceeds 0.9 for most attributes at higher slider scales, e.g., *age* 0.92 (SDXL) and 0.96 (FLUX), *curly hair* 0.95 (SDXL) and 0.87 (FLUX), *smile* 0.93 (SDXL) and 0.92 (FLUX) at scale 3, meaning that the bulk of the attribute-aligned CLIP shift lands inside the targeted mask. $\text{LPIPS}_{\text{LOC}}$ ratios are strongly positive on both backbones, with FLUX exhibiting in particular very large inside/outside contrasts (e.g. *curly hair* reaches $\text{LPIPS}_{\text{LOC}} = 32.1$ at scale 3); this is a consequence of the area-normalized ratio when the outside-mask perceptual change is very small, as is desired. The mechanism transfers cleanly across backbones with no architecture-specific adjustment beyond the resolution matching of the mask (Sec. 3.1). At the smallest slider scale on SDXL, two

style-like attributes, *day/night* and *painterly*, score below the strong regime, with $\text{CLIP}_{\text{LOC}} = 0.030$ on *day/night* and 0.481 on *painterly*. This is not a limitation of the mechanism but a property of those attributes at scale 1: a single unit of slider strength is simply not enough to register a visible style shift on the gray-filled crop, and the scores recover as soon as the scale grows (*day/night* 0.869 and 0.924 at scales 2 and 3; *painterly* 0.801 and 0.870).

Takeaway. Overall the mask-guided application is stable: the localization scores are consistently high across attributes, scales, and both backbones. Much of this robustness comes from the per-step blend itself (Sec. 4): re-applying the mask at every denoising step, rather than compositing regions once at the end, keeps the two sides of the mask boundary coherent and free of seam artefacts.

6. Conclusions

LoRA adapters have become a standard deep-learning tool; in the case of diffusion models, a small, separately trained module is enough to steer what a fixed generator produces. That control, however, is global. In this work we set out to give it a spatial dimension, confining the effect of a LoRA adapter to a chosen subject or region of a multi-subject scene. We proposed a mask-guided application that does so at inference time: a globally-trained adapter is gated on an externally-provided segmentation mask through a per-step blend of the masked and unmasked predictions. To assess the mechanism critically, we adapted two standard metrics, LPIPS and CLIP, into region-aware variants that measure not just how much an image changes but where the change lands. A central finding is that the mechanism is genuinely architecture-agnostic. Because the blend acts on the model’s per-step prediction rather than on any architectural internal, the same operator applies to two structurally different generators: the UNet noise predictor ε_θ of a latent diffusion model (SDXL) and the MM-DiT velocity field v_θ of a rectified-flow transformer (FLUX). The only backbone-specific element is matching the mask resolution to each model’s prediction grid, a 128×128 latent for SDXL against a 64×64 patch grid for FLUX. Equally important is that the blend is re-applied at *every* denoising step rather than compositing regions once at the end: since the masked and unmasked forwards start from the same latent at each step, the iterative sampler keeps the two sides of the mask boundary coherent on its own, with no seam artefacts and without any dedicated boundary treatment. Empirically the mask-guided application proved precise and robust, with consistently high localization scores across attributes, scales, and both backbones. We stress-tested it on Concept Sliders, a deliberately hard case because each slider encodes a fine-grained semantic direction rather than a whole subject; the same mechanism applies unchanged to other scaled LoRA adapters, such as identity or subject

adapters. We see the mask-guided application above all as a practical editing tool: it fits a creative workflow in which the user wants finer control over a generation, applying a localized change without disturbing the rest of the scene, precisely what a prompt alone cannot achieve.

References

- [1] Black Forest Labs. FLUX.1: A suite of open rectified-flow text-to-image models. Model release, August 2024, 2024. <https://huggingface.co/black-forest-labs/FLUX.1-dev>. 2, 4
- [2] Guillaume Couairon, Jakob Verbeek, Holger Schwenk, and Matthieu Cord. DiffEdit: Diffusion-based semantic image editing with mask guidance. In *International Conference on Learning Representations (ICLR)*, 2023. <https://arxiv.org/abs/2210.11427>. 2
- [3] Yusuf Dalva, Hidir Yesiltepe, and Pinar Yanardag. LoRASHop: Training-free multi-concept image generation and editing with rectified flow transformers. *arXiv preprint arXiv:2505.23758*, 2025. <https://arxiv.org/abs/2505.23758>. 2, 3
- [4] Rohit Gandikota, Joanna Materzyńska, Tingrui Zhou, Antonio Torralba, and David Bau. Concept sliders: LoRA adaptors for precise control in diffusion models. In *European Conference on Computer Vision (ECCV)*, 2024. <https://arxiv.org/abs/2311.12092>. 1, 2, 4
- [5] Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Prompt-to-prompt image editing with cross attention control. In *International Conference on Learning Representations (ICLR)*, 2023. <https://arxiv.org/abs/2208.01626>. 2
- [6] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. <https://arxiv.org/abs/2006.11239>. 2
- [7] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. <https://arxiv.org/abs/2106.09685>. 1, 2
- [8] Edo Kadosh et al. Tight inversion: Image-conditioned inversion for real image editing. *arXiv preprint arXiv:2502.20376*, 2025. <https://arxiv.org/abs/2502.20376>. 1
- [9] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. <https://arxiv.org/abs/2304.02643>. 2, 4
- [10] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023. <https://arxiv.org/abs/2209.03003>. 2
- [11] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. SDXL: Improving latent diffusion models for high-resolution image synthesis. In *International Conference on Learning Representations (ICLR)*, 2024. <https://arxiv.org/abs/2307.01952>. 2, 4
- [12] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021. <https://arxiv.org/abs/2103.00020>. 4
- [13] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. DreamBooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. <https://arxiv.org/abs/2208.12242>. 2
- [14] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. <https://arxiv.org/abs/1801.03924>. 4

A. External Mask-Guided: quantitative results

Concept	Scale 1		Scale 2		Scale 3	
	LPIPS _{LOC} ↑	CLIP _{LOC} ↑	LPIPS _{LOC} ↑	CLIP _{LOC} ↑	LPIPS _{LOC} ↑	CLIP _{LOC} ↑
Age	5.66	0.871	7.06	0.921	6.97	0.923
Curly Hair	6.75	0.929	6.82	0.947	7.04	0.951
Day/Night	1.98	0.030	2.67	0.869	3.35	0.924
Fur Length	4.15	0.670	4.41	0.792	5.01	0.815
Painterly	2.71	0.481	3.56	0.801	3.66	0.870
Smile	4.70	0.870	4.88	0.932	4.65	0.934

Table 1. External Mask-Guided on SDXL: median LPIPS_{LOC} and CLIP_{LOC} over 120 multi-subject scenes (6 concepts × 20 scenes). The mask is applied to one subject. Bold marks the best scale per row and metric.

Concept	Scale 1		Scale 2		Scale 3	
	LPIPS _{LOC} ↑	CLIP _{LOC} ↑	LPIPS _{LOC} ↑	CLIP _{LOC} ↑	LPIPS _{LOC} ↑	CLIP _{LOC} ↑
Age	8.66	0.672	9.45	0.925	9.61	0.955
Curly Hair	23.68	0.834	25.86	0.842	32.12	0.868
Fur Length	10.82	0.728	11.90	0.823	11.76	0.846
Painterly	3.44	0.624	3.80	0.826	4.36	0.831
Smile	7.44	0.907	5.82	0.899	6.66	0.924

Table 2. External Mask-Guided on FLUX.1-dev: median LPIPS_{LOC} and CLIP_{LOC} over 100 multi-subject scenes (5 concepts × 20 scenes). The mask is applied to one subject. Bold marks the best scale per row and metric.

B. Image examples: baseline

Qualitative examples of the baseline: a vanilla Concept Slider LoRA applied to a multi-subject scene, without our localization mechanism. The slider’s effect spreads across every subject in the scene that matches the attribute.



Figure 2. Baseline: the *age* slider applied globally, with no localization; every person in the scene ages.



Figure 3. Baseline: the *painterly* style slider applied globally, with no localization; the whole scene is restyled.

C. Image examples: mask-guided

Additional qualitative examples of the external mask-guided application. Each row shows, from left to right: the unedited generation, the SAM mask (shown in green), and the localized edit(s).



Figure 4. Mask-guided *age* slider on FLUX.1-dev, scene 1.



Figure 5. Mask-guided *age* slider on FLUX.1-dev, scene 2.



Figure 6. Mask-guided *smile* slider on FLUX.1-dev.



Curly Hair on Mask

Figure 7. Mask-guided *curly hair* slider on FLUX.1-dev, scene 1.



Curly Hair on Mask

Figure 8. Mask-guided *curly hair* slider on FLUX.1-dev, scene 2.



Painting Style on Mask

Figure 9. Mask-guided *painterly* style slider on FLUX.1-dev.



Painting Style on Mask

Figure 10. Mask-guided *painterly* style slider on SDXL, scene 1.



Painting Style on Mask

Figure 11. Mask-guided *painterly* style slider on SDXL, scene 2.



Long Fur on Mask

Figure 12. Mask-guided *fur length* slider on SDXL.



Night Sky on Mask



Figure 13. Mask-guided *day/night* slider on SDXL.

D. Internal masks extracted via the LoRAShop adaptation



Figure 14. Internal cross-attention mask from our LoRAShop adaptation, intended to isolate the man so that the *curly hair* slider edits him alone. The mask is too coarse: it also covers the woman, and the slider’s effect leaks onto her. This is what motivates the external-mask design of Sec. 4.